

🐼 3D Coordinates Extension — Build the Panda (Intermediate)

Topic: 3D Panda (x, y, z) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Lesson 6) ·

Time: about 40 minutes

Build a **standing panda** in 3D — white body, black ears, black eye patches, black legs. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

● Stand on your **home spot** (red block) every run. Move your feet, move your panda.

🎨 White parts = **white concrete**. Black parts = **black wool**.

1 · Predict 🧠

Read each set. Write a sentence for each.

```
place black block at (2, 3, 1)
place black block at (2, 4, 1)
place black block at (2, 5, 1)
```

Only y changes. What happens to this leg?


```
place white block at (2, 8, 1)
place white block at (2, 8, 2)
place white block at (2, 8, 3)
```

Only z changes. What happens to the line?


```
place white block at (2, 7, 1)
place black block at (2, 7, 1)
```

Same (x, y, z). What do you see, and why?

2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

Bug A — This ear should sit one step **deeper**, at (2, 9, 2). It has only two numbers, so it lands flat on the front.

```
place black block at (2, 9)
```

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences)

Bug B — This eye should be **up high on the front**: (5, 7, 1). The last two numbers are swapped.

```
place black block at (5, 1, 7)
```

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences)

3 · Show Your Work 📷 🎥

Look at the 🐼 panda in the world. Walk around it. Then build **one full panda** on your home spot — head, two ears, two eyes, two legs.

Front face (keep $z = 1$): fill a white head from (2, 5, 1) to (5, 8, 1). Add two black ears on top — (2, 9, 1) and (5, 9, 1). Add two black eyes — (2, 7, 1) and (5, 7, 1). Add two black legs at the bottom — fill (2, 3, 1) to (2, 4, 1) and (5, 3, 1) to (5, 4, 1).

Add depth: build the white head again at $z = 2$, then $z = 3$, until your panda is as deep as the real one. Build the ears and legs deeper too.

Face on front only: at $z = 2$ and $z = 3$, place plain white where the eyes were.

💡 Tip: the eyes only show on the **front**. The sides and back are plain white.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera at the panda. Name the parts.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your **one** video to teacher on KakaoTalk.